To get the monthly sale values, it is required to get the *OrderDate* based grouping of records. The month-wise sale values can be computed by summing the product of *unitPrice* and *Quantity* of each month of each year.

```
SELECT year
, SUM(CASE WHEN month = 1 THEN unitPrice*Quantity END) Jan
, SUM(CASE WHEN month = 2 THEN unitPrice*Quantity END) Feb
, SUM(CASE WHEN month = 3 THEN unitPrice*Quantity END) Mar
, SUM(CASE WHEN month = 4 THEN unitPrice*Quantity END) Apr
, SUM(CASE WHEN month = 5 THEN unitPrice*Quantity END) May
, SUM(CASE WHEN month = 6 THEN unitPrice*Quantity END) Jun
, SUM(CASE WHEN month = 7 THEN unitPrice*Quantity END) Jul
, SUM(CASE WHEN month = 8 THEN unitPrice*Quantity END) Aug
, SUM(CASE WHEN month = 9 THEN unitPrice*Quantity END) Sep
, SUM(CASE WHEN month = 10 THEN unitPrice*Quantity END) Oct
, SUM(CASE WHEN month = 11 THEN unitPrice*Quantity END) Nov
, SUM(CASE WHEN month = 12 THEN unitPrice*Quantity END) Dec
FROM (SELECT Invoices.*
, EXTRACT(YEAR FROM OrderDate) year
, EXTRACT(MONTH FROM OrderDate) month
FROM Invoices
) Invoices
GROUP BY year
ORDER BY year
```

Year	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
1996							30192.1	26609.4	27636	41203.6	49704	50953.4
1997	66692.8	41207.2	39979.9	55699.39	56823.7	39088	55464.93	49981.69	59733.02	70328.5	45913.36	77476.26
1998	100854.7	104562	109825.5	134630.6	19898.66							

Similar to the last pivot table, this query will produce a report of year-wise monthly sales, as given above. Further statistical analysis may be used for high level analysis of revenue:

```
SELECT
STD(jan_sales), STD(feb_sales), STD(mar_sales), STD(apr_
sales), STD(may_sales), STD(jun_sales), STD(jul_sales),
STD(aug_sales), STD(sep_sales), STD(oct_sales), STD(nov_
sales), STD(dec_sales)
FROM (SELECT year
, SUM(CASE WHEN month = 1 THEN unitPrice*Quantity END) jan_
sales
, SUM(CASE WHEN month = 2 THEN unitPrice*Quantity END) feb_
sales
…..
……
…….. 
FROM (SELECT Invoices.*
, EXTRACT(YEAR FROM OrderDate) year
, EXTRACT(MONTH FROM OrderDate) month
FROM Invoices
) Invoices
GROUP BY year) AS stat
```

Readers are encouraged to explore pivot tables with other high level data analysis techniques in SQL. END

By: Dipankar Ray

The author is a member of IEEE, IET (UK) and has more than 20 years of experience in open source versions of UNIX operating systems.